

Quantitative Thinking #2

How Much Would You Weigh on Earth, Moon, and Mars?

Date

August 2026

Project Goal

I wanted to understand how gravity affects weight on different worlds and why astronauts experience different conditions on the Moon and Mars compared to Earth.

What I Worked On

- Researched the concepts of mass, weight, and gravity.
- Learned how gravity differs on Earth, the Moon, and Mars.
- Studied how planetary environments affect weight.
- Created an Excel spreadsheet to compare weights under different gravitational conditions.
- Calculated weight values using different gravity values.
- Organized the results into a data table.
- Created a graph to visualize the relationship between gravity and weight.

What I Learned

Before starting this analysis, I often used the words mass and weight as if they meant the same thing.

However, I learned that mass remains constant while weight changes depending on gravity.

I was surprised to see how much lower weight becomes on the Moon compared to Earth. Even though a person's mass stays exactly the same, the weaker gravity significantly reduces weight.

This analysis helped me better understand how gravity affects movement and why astronauts experience different conditions on different worlds.

Results

- Weight was highest on Earth.
- Weight was lowest on the Moon.
- Mars produced values between Earth and the Moon.
- Weight increased as mass increased on all three worlds.
- The graph clearly showed how gravity influences weight.

Challenges

One of the most interesting parts of this project was understanding the difference between mass and weight.

At first, I assumed they referred to the same thing. However, while working on the calculations, I discovered that mass remains constant while weight depends on gravity.

Key Lessons Learned

- Mass and weight are not the same.
- Gravity directly affects weight.
- The same object can behave differently on different worlds.
- Mathematics can be used to explain space science concepts.
- Data visualization helps make scientific relationships easier to understand.

Final Reflection

This analysis started with a simple question: How much would I weigh on Earth, the Moon, and Mars?

Although the calculations were straightforward, the results helped me better understand one of the most interesting concepts in space science.

What interested me most was seeing how the same person can have different weights on different worlds while their mass remains exactly the same. This project showed me how mathematics can be used to explain real-world space science concepts and how simple calculations can reveal interesting scientific insights.

Mass stays the same. Weight changes.